Mayank Chandak

(412) 287-3522 | mayank.chandak21@gmail.com | mayankchandak.vercel.app | github.com/mnm-21 | linkedin.com/in/mayank-chandak

Education

Carnegie Mellon University **Aug. 2026 – May 2028**
Master of Science in Robotic Systems Development *Pittsburgh, PA*
Relevant Coursework: Advanced Computer Vision, Robot Mobility, Robot Autonomy

Indian Institute of Technology Madras **Jul. 2022 – May 2026**
Bachelor of Technology in Mechanical Engineering, Minor in Artificial Intelligence and Machine Learning *Chennai, India*
CGPA: **9.38/10** Top 5/205 Mechanical Engineering

Experience

Robotics Capstone, Nissan-Sponsored Project **Aug. 2026 – Present**
Vision-Based Learning for Humanoid Manipulation *Carnegie Mellon University, Pittsburgh, PA*

- Architecting an end-to-end vision-to-action pipeline for a Unitree G1 humanoid, evaluating open-source VLAs including GR00T and π0.5 for fine-tuning on real-world Nissan manufacturing tasks.
- Designing a human-video-to-robot learning pipeline under limited demonstrations, exploring multi-view camera configurations, SAM 3D reconstruction, human-to-robot correspondence, and deployment-oriented policy learning.

Undergraduate Researcher, INSPIRE Lab **May 2025 – May 2026**
Surgical Phase-Aware Reinforcement Learning *IIT Madras, Chennai, India*

- Developed a phase-aware PPO framework for 4-DOF autonomous laparoscopic camera control, adapting centering, zoom, and motion objectives across surgical phases.
- Built a SOFA/LapGym simulation with deformable tissue, articulated instruments, FEM collision modeling, and pivotized endoscope control under RCM constraints; achieved **40% higher centering** and **67% lower motion jitter** than phase-blind RL.

Young Research Fellow, Computational Imaging Lab **Jul. 2024 – Mar. 2025**
FlareGS: 4D Flare Removal for Urban Scenes *IIT Madras, Chennai, India*

- Built the physics-based synthetic flare generation pipeline for FlareGS, contributing to a **1,000-pair** dataset across three controlled flare-severity regimes using real and synthetic urban driving scenes.
- Set up the end-to-end reconstruction pipeline, including **RGB/depth fusion, UFormer restoration, Gaussian initialization, camera-pose configuration, optimization losses, and 4D Gaussian Splatting training** for temporally consistent novel-view synthesis.
- Achieved **24.83/24.46 dB masked PSNR** at M2/M3 flare severity, improving over UFormer by **1.26/1.85 dB**; also contributed to VidFlareNet, reaching **20.03 PSNR** and **8.24 temporal-consistency loss**.

ML Engineering Intern, Quantiphi **Jun. 2024 – Jul. 2024**
Generative AI and Document Intelligence *Mumbai, India*

- Benchmarked Sparrow Parse and internal document-intelligence models for structured extraction from tabulated client documents, comparing model outputs across document-processing workflows.

Publications and Submissions

M. Chandak*, S. S. T. Kuppa*, R. G. R. Matta, V. Gupta, K. Mitra, "FlareGS: 4D Flare Removal using Gaussian Splatting for Urban Scenes," *ICCV 2025 Workshop (2COOOL)*, 2025. *Equal contribution.

M. Chandak, S. Jha, V. Menagar, N. Patel, "Phase-Belief-Conditioned Reinforcement Learning for Autonomous Laparoscopic Camera Control," *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. **Under review**.

Projects

SONIC Whole-Body Teleoperation on Unitree G1 *Oct. 2026*

- Deployed NVIDIA GR00T SONIC for Quest 3S teleoperation on a Unitree G1, with 500 Hz DDS motor commands and Dex3-1 trigger grasp control; brought up the runtime on a second G1's onboard Jetson Orin NX.
- Built neck-mounted XR pose retargeting and body-follow locomotion; Ruckig smoothing reduced scripted target-acceleration RMS by **10**×. Implemented safety checks and controller handover measured at **145 ms** across 4/4 real-robot runs; backed by 150 tests.

MyoFootball: Reward-Driven Musculoskeletal Control for Goal-Oriented Locomotion

- Extended **MyoSuite's musculoskeletal MyoLeg environment** with ball dynamics and contact mechanics; designed a multi-stage reward hierarchy for locomotion, kicking, and balance.
- Iteratively tuned the reward formulation across randomized ball placements, achieving **80% goal-scoring accuracy** through extensive reward-design experiments.

Technical Skills

Programming	Python, C++, MATLAB, Bash/zsh
Robotics & Simulation	ROS 2, MuJoCo, SOFA, LapGym, Robosuite, MyoSuite, CoppeliaSim
AI/ML	PyTorch, TensorFlow, Stable-Baselines3, Reinforcement Learning, Deep Learning, Computer Vision
Tools	Git, GitHub, Linux/Ubuntu, OpenCV, NumPy, Pandas, SolidWorks, LaTeX

Honors & Achievements

- **Young Research Fellow Award**, IIT Madras: selected 1 of 17 students from 300+ applicants for funded undergraduate research.
- **Top 5% JEE Advanced** — **Top 1% JEE Main**.